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CS11021– Programming Laboratory (Intake 43)

Lab Sheet 14

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1. You are given two matrices:

   Matrix A (2×3):

   ```
   1  2  3
   4  5  6
   ```

   Matrix B (3×2):

   ```
   7   8
   9   10
   11  12
   ```

   Task:
   Write a C++ program to:

   a. Store both matrices using 2D arrays
   b. Multiply Matrix A and Matrix B
   c. Display the resultant matrix

2. A supermarket sells 3 products in 2 different branches.

   - Matrix A represents **quantity sold**:

   ```
   Branch1:  10   5   8
   Branch2:  7    6   9
   ```

- Matrix B represents **price per product (column matrix)**:

```
2
4
3
```

Task:

i. Write a C++ program to calculate total revenue for each branch using matrix multiplication
ii. Display the total revenue of each branch

3. You are given the following matrix representing connections between 4 nodes:

```
0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 0
```

Task:

a. Identify whether this matrix is an adjacency matrix
b. Write a C++ program to:

   i. Store the matrix
   ii. Display all connected node pairs

4. A small city has 4 locations: A, B, C, D. The connectivity between locations is given as:

```
0 1 1 0
1 0 1 1
1 1 0 0
0 1 0 0
```

Task:

a. Represent this using an adjacency matrix in C++
b. Write a program to:

   - Display all directly connected locations
   - Count the number of connections for each location

5. A company has 3 departments exchanging resources. The interaction is given by:

   Adjacency Matrix:

   ```
   0 1 1
   1 0 1
   1 1 0
   ```

   Resource Transfer Matrix:

   ```
   5 2 3
   4 6 1
   7 2 8
   ```

   Task:

   a. Write a C++ program to:
      i. Multiply the adjacency matrix with the resource matrix
   b. Interpret the result as total resource flow between departments
   c. Display the final matrix and explain what each value represents